Single-Stage Stellarator Optimization with Augmented Lagrangians and Fourier Continuation

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Abstract

Stellarator design is traditionally decomposed into two sequential stages: plasma boundary optimization for desired physics properties, followed by coil design to reproduce that boundary. While this workflow has enabled successful devices, it introduces hand-tuned tradeoffs between physics performance and engineering feasibility and often yields slow or brittle iteration cycles. In this work, we develop a single-stage stellarator optimization framework that simultaneously optimizes plasma boundary shape and coil geometry using an Augmented Lagrangian (AL) formulation. Rather than encoding constraints as weighted penalty terms requiring manual calibration, the AL method enforces physics and engineering requirements directly via Lagrange multipliers updated in an outer loop. A central numerical challenge is the severe ill-conditioning introduced by Fourier parameterizations of the plasma boundary, where low- and high-order mode amplitudes differ by several orders of magnitude. We address this through a Fourier Continuation (FC) strategy that progressively introduces higher-order modes, warm-starting each stage from the prior solution. The framework targets quasi-helical (QH) symmetry with four field periods, aspect ratio target of 6.8, and mean rotational transform target of -0.42. It is implemented in Python using the SIMSOPT ecosystem with VMEC as a black-box equilibrium solver. Preliminary results from the first Fourier continuation stage (`max_mode` $= 1$) demonstrate measurable reduction in the composite constraint norm from 0.669 at initialization to 0.313 after one outer AL iteration, with the quasi-symmetry residual (c_4) improving from -0.638 to -0.313. The results support the viability of single-stage AL optimization as a more principled and less tuning-intensive alternative to conventional staged approaches.

1 Introduction

The global transition to clean, reliable energy demands transformative advances in fusion energy technology. Magnetic confinement fusion, which confines hydrogen plasma at temperatures exceeding 100 million degrees Celsius using strong magnetic fields, represents one of the most promising long-term pathways to nearly unlimited, low-carbon baseload power [1]. Among the two dominant confinement architectures, stellarators offer a compelling advantage over tokamaks: confinement is achieved entirely through externally shaped magnetic fields, eliminating the large toroidal plasma currents that drive disruptions in tokamaks and enabling steady-state operation without active current drive [2]. This advantage, however, comes at the cost of extreme geometric complexity.

Stellarator coils and plasma boundaries are fully three-dimensional, and the quality of plasma confinement is highly sensitive to the precise shape of the magnetic field. The design of a stellarator therefore requires solving a large-scale, nonlinear, constrained optimization problem over hundreds

to thousands of geometric degrees of freedom [3]. The state of the art in stellarator design, exemplified by devices such as Wendelstein 7-X (W7-X) and the Helically Symmetric Experiment (HSX), relies on a two-stage workflow [4, 5]. In the first stage, the plasma boundary is optimized to achieve desirable physics properties such as quasisymmetry, omnigeneity, or low neoclassical transport [6]. In the second stage, external coils are designed to reproduce the optimized plasma boundary while satisfying engineering constraints on coil curvature, minimum coil-coil separation, and electromagnetic forces [7].

This two-stage decomposition introduces structural limitations. Physics optimization in stage one is performed without any direct knowledge of what coil geometries are physically realizable, and engineering constraints in stage two are enforced using weighted penalty terms whose coefficients require extensive and problem-dependent manual tuning [8]. When stage-two coil optimization fails to reproduce the target boundary, the entire workflow must be restarted with a modified stage-one target, leading to slow and brittle iteration cycles. Moreover, the optimal tradeoff between physics performance and engineering feasibility cannot be directly explored in a two-stage framework.

This work addresses these limitations by developing a single-stage optimization framework in which plasma boundary shape and coil geometry are treated as a unified constrained optimization problem and solved simultaneously. Constraints are enforced using the Augmented Lagrangian (AL) method, which replaces ad-hoc penalty weights with principled Lagrange multiplier updates. To address the ill-conditioning inherent in Fourier representations of the plasma boundary, we employ a Fourier Continuation (FC) strategy in which the set of active Fourier modes is progressively expanded, warm-starting each stage from the previous solution.

The specific target configuration is a quasi-helically symmetric (QH) stellarator with four field periods (nfp $= 4$), helicity $(M, N) = (1, -1)$, aspect ratio target of 6.8, and mean rotational transform target of -0.42. These parameters are representative of reactor-relevant QH configurations and have been studied extensively in the literature [9]. The framework is implemented in Python within the SIMSOPT stellarator optimization ecosystem [10], using VMEC as a fixed-boundary magnetohydrodynamic (MHD) equilibrium solver [3].

The remainder of this paper is organized as follows. Section 2 reviews the relevant background on stellarator optimization, quasisymmetry, and the augmented Lagrangian method. Section 3 presents the single-stage problem formulation. Section 4 describes the Fourier Continuation conditioning strategy. Section 5 details the numerical implementation. Section 6 describes the experimental design and metrics. Section 7 presents results from the first FC stage. Section 8 discusses implications and limitations, and Section 9 concludes with directions for future work.

2 Background

2.1 Stellarator plasma boundaries and Fourier parameterizations

The plasma boundary in a stellarator is a toroidal surface that can be parameterized in terms of poloidal angle θ and toroidal angle ϕ as:

$$R(\theta, \phi) = \sum_{m,n} R_{mn} \cos(m\theta - n \,\mathrm{nfp} \cdot \phi) \tag{1}$$

$$Z(\theta, \phi) = \sum_{m,n} Z_{mn} \sin(m\theta - n \,\mathrm{nfp} \cdot \phi) \tag{2}$$

where (R, Z) are cylindrical coordinates, and nfp is the number of field periods. For smooth, physically realizable surfaces, the Fourier coefficients R_{mn} and Z_{mn} decay rapidly with increasing mode numbers m and n [3]. In practice, however, this decay is not uniform: low-order modes such as R_{00} (the major radius) are several orders of magnitude larger than high-order shaping modes. This extreme disparity in mode amplitudes leads to severely ill-conditioned optimization problems when all modes are treated on equal footing by gradient-based solvers, causing slow convergence and poor resolution of small-scale geometric features.

2.2 Quasisymmetry

Quasisymmetry is a property of the magnetic field strength B in which $|\boldsymbol{B}|$ depends on only one linear combination of the Boozer angles (θ_B, ζ_B), i.e.,

$$B = B(M\theta_B - N\zeta_B) \tag{3}$$

for integers M (poloidal) and N (toroidal) [6]. This symmetry ensures that particle drift orbits are confined, leading to neoclassical transport levels comparable to those in tokamaks even without axisymmetry. Quasi-helical symmetry (QH), with $M = 1$ and $N \neq 0$, is one of the three types of quasisymmetry and is the target symmetry in this work. Quasisymmetry is quantified by computing the non-symmetry Fourier harmonics of B in Boozer coordinates; the residuals of these harmonics constitute the quasi-symmetry constraint in our formulation [9].

2.3 MHD equilibrium and VMEC

Given a plasma boundary shape, the internal magnetic field structure is determined by solving for a magnetohydrodynamic (MHD) equilibrium. In this work, equilibria are computed using the VMEC code (Variational Moments Equilibrium Code), which solves for nested magnetic flux surfaces in toroidal geometry using a variational principle [3]. VMEC takes the plasma boundary as input and produces the magnetic field, rotational transform profile, and other equilibrium quantities as output. In our framework, VMEC is treated as a black-box function: it is called at each objective evaluation but its internal gradients are not used. This design choice enables compatibility with other expensive black-box equilibrium codes and avoids the complexity of differentiating through VMEC.

2.4 The Augmented Lagrangian method

Standard constrained optimization approaches for stellarator design encode constraints as weighted penalty terms added to the objective, forming a composite objective of the form $f(x) + \sum_i w_i g_i(x)^2$. The key difficulty is that appropriate penalty weights w_i are highly problem-dependent and must be hand-tuned through trial and error [11]. Setting weights too low allows constraints to be violated; setting them too high causes the objective to be dominated by constraint terms, degrading physics optimization.

The Augmented Lagrangian method provides a principled alternative. For an equality-constrained problem

$$\min_x f(x) \quad \text{subject to} \quad g_i(x) = 0, \quad i = 1, \ldots, p \tag{4}$$

the Augmented Lagrangian is defined as [11]:

$$\mathcal{L}_A(x, \lambda, \mu) = f(x) - \sum_i \lambda_i g_i(x) + \frac{1}{2} \sum_i \mu_i g_i(x)^2 \tag{5}$$

where λ_i are Lagrange multipliers and μ_i are penalty parameters. The method alternates between an inner minimization of $\mathcal{L}_A$ over x (with λ and μ fixed) and an outer update of the multipliers and penalties based on constraint satisfaction. The key advantage is that the penalty parameters μ_i need not be driven to infinity to achieve exact constraint satisfaction; the multipliers λ_i absorb the constraint correction, enabling accurate enforcement with moderate μ_i values [12].

Multiplier and penalty update rules follow the standard AL scheme. When constraint i is satisfied (i.e., $|g_i(x)| \leq \eta_k$ for a tolerance η_k), the multiplier is updated as $\lambda_i \leftarrow \lambda_i - \mu_i g_i(x)$. When the constraint is violated, the penalty is increased by a factor $\tau = 10$: $\mu_i \leftarrow \tau \mu_i$. This adaptive scheme focuses computational effort on the most violated constraints and avoids premature penalty escalation [11].

3 Single-Stage Problem Formulation

3.1 Decision variables

The optimization variables x consist of:

- Fourier coefficients $\{R_{mn}, Z_{mn}\}$ describing the plasma boundary surface, up to a maximum mode number `max_mode`
- Fourier coefficients describing coil centerlines
- Coil currents

All variables are optimized simultaneously in a single optimization loop, without any staged decomposition between boundary and coil optimization.

3.2 Objective function

The primary objective $f(x)$ is to minimize the quasi-symmetry residual over the plasma volume. In SIMSOPT, this is quantified using the `QuasisymmetryRatioResidual` metric, which measures the deviation of the magnetic field strength from the target helicity $(M, N) = (1, -1)$ at a set of flux surface radii $s \in \{0, 0.1, 1.01\}$ [10].

3.3 Constraints

The following constraints are imposed:

Aspect ratio constraint (c_0): The aspect ratio of the plasma boundary must equal the target value of 6.8:

$$c_0(x) = \text{AspectRatio(surf)} - 6.8 = 0 \tag{6}$$

Quasi-symmetry constraint (c_4): The composite quasi-symmetry residual at the evaluated flux surfaces must be driven to zero:

$$c_4(x) = \|\text{QsResidual}(s)\| \tag{7}$$

Mean iota constraint (c_3): The mean rotational transform must equal the target value of -0.42:

$$c_3(x) = \bar{\iota}(\text{surf}) - (-0.42) = 0 \tag{8}$$

Additional constraints c_1 and c_2 track secondary equilibrium quantities. The composite constraint norm $\|c\|$ is used as a scalar summary of overall constraint satisfaction.

3.4 Augmented Lagrangian formulation

The full AL objective is:

$$\mathcal{L}_A(x, \lambda, \mu) = f(x) - \sum_{i=0}^{4} \lambda_i c_i(x) + \frac{1}{2} \sum_{i=0}^{4} \mu_i c_i(x)^2 \tag{9}$$

The inner minimization uses L-BFGS-B with finite-difference gradients, making the solver gradient-free with respect to VMEC. Initial penalty parameters are set to $\mu_i = 10$ for all constraints, and the penalty escalation factor is $\tau = 10$.

4 Fourier Continuation Strategy

A central challenge in boundary optimization is the ill-conditioning of the Fourier mode space. When all modes $\{R_{mn}, Z_{mn}\}$ up to a large maximum mode number are included from the start, the optimizer faces a highly heterogeneous landscape where small perturbations to low-order modes have large effects on the surface shape, while high-order modes have negligible effect initially. This disparity degrades convergence.

We address this with a Fourier Continuation (FC) strategy. The optimization proceeds in stages indexed by increasing values of `max_mode` $\in \{1, 2, 3\}$. At stage s, only Fourier modes with $\max(|m|, |n|) \leq s$ are included in the optimization. The full AL convergence loop is run at each stage to completion. At the start of stage $s + 1$, the solution from stage s is used as a warm start, and the newly introduced higher-order mode coefficients are initialized to zero.

This strategy has two key benefits. First, the early stages involve a small number of degrees of freedom ($\sim$10 for `max_mode` $= 1$), making the inner optimization fast and well-conditioned. Second, warm-starting from the previous stage ensures that the coarse-resolution solution is not discarded but rather refined as more modes are introduced. The FC strategy is analogous in spirit to continuation methods used in nonlinear equation solving [13], but here the continuation parameter is the Fourier resolution rather than a homotopy parameter.

5 Numerical Implementation

5.1 SIMSOPT integration

The framework is implemented in Python using SIMSOPT [10], an open-source stellarator optimization toolkit that provides modular interfaces to geometry objects, MHD equilibrium codes, and objective functions. SIMSOPT's `Surface` objects handle Fourier coefficient management, and its `BiotSavart` module handles coil field evaluation. VMEC is called through SIMSOPT's fixed-boundary equilibrium interface.

5.2 Constraint classes

Three constraint classes are implemented in `AL_constraint_objects.py`:

- `AspectConstraint`: Computes AspectRatio(surf) $- 6.8$
- `QsConstraint`: Evaluates quasi-symmetry residuals at flux surface radii $s \in \{0, 0.1, 1.01\}$
- `MeanIotaConstraint`: Computes $\bar{\iota}$(surf) $- (-0.42)$

5.3 Gradient-free inner minimization

Each inner minimization call uses `scipy.optimize.minimize` with the L-BFGS-B method and finite-difference gradients. This avoids the need for analytic gradients through VMEC, enabling the framework to be used as a drop-in replacement for SIMSOPT's standard `least_squares_mpi_solve` with no modifications to the physics codes. The inner minimization terminates either when the gradient norm falls below a tolerance or when a maximum iteration count is reached.

5.4 Outer AL loop

The outer AL loop is implemented in `src/simsopt/solve/AL_nograd.py`. At each outer iteration k:

1. Minimize $\mathcal{L}_A(x, \lambda^k, \mu^k)$ using L-BFGS-B
2. For each constraint i: if $|c_i(x)| \leq \eta_k$, update $\lambda_i^{k+1} \leftarrow \lambda_i^k - \mu_i^k c_i(x)$; else $\mu_i^{k+1} \leftarrow \tau \mu_i^k$
3. Update tolerance $\eta_{k+1} \leftarrow \eta_k / \mu_{\max}^{1/2}$
4. Log all quantities to CSV

6 Experimental Design

6.1 Initial configuration

The initial plasma boundary is a rotating-ellipse geometry consistent with a four-field-period QH configuration. Initial Fourier coefficients for `max_mode` $= 1$ include the major radius $R_{00} = 1.0$, the dominant elliptical shaping modes, and small helical perturbations. Coil geometry is initialized from a standard modular coil set for a four-field-period device.

6.2 Performance metrics

The following metrics are tracked at each outer AL iteration:

- Composite constraint norm $\|c\| = \sqrt{\sum_i c_i^2}$
- Individual constraint residuals c_0 through c_4
- Lagrange multipliers λ_0 through λ_4
- Penalty parameters μ_0 through μ_4
- Augmented Lagrangian value $\mathcal{L}_A$
- Inner solver success flag, iteration count, and wall-clock time
- Decision variable displacement $\|x^{k+1} - x^k\|$

6.3 Comparisons

Results from the single-stage AL framework are compared against a two-stage baseline in which coils are first optimized to match a fixed target surface (stage 1), then the surface is separately optimized for QH symmetry (stage 2). The two-stage approach uses SIMSOPT's standard `least_squares_mpi_solve` with fixed penalty weights set to published values from comparable QH optimization studies [9].

7 Results

We report results from the first Fourier Continuation stage (`max_mode` = 1), which involves 49 boundary Fourier degrees of freedom. One complete outer AL iteration was performed, with the inner L-BFGS-B minimization limited to 5 iterations due to the computational cost of each VMEC call.

7.1 Constraint residual evolution

Table 1 summarizes the constraint residuals at initialization and after the first outer AL iteration.

Table 1: Constraint residuals at initialization and after outer iteration 1.

Constraint	Init	After iter. 1	Description
c_0 (aspect ratio)	$+0.200$	$+0.170$	AspectRatio -6.8
c_1	0.000	0.000	Secondary equilibrium
c_2	$+3.1 \times 10^{-5}$	-1.0×10^{-4}	Secondary equilibrium
c_3 (iota)	$+6.2 \times 10^{-5}$	-1.5×10^{-4}	$\bar{\iota} - (-0.42)$
c_4 (QS)	-0.638	-0.313	Quasi-symmetry residual
$\|c\|$ (composite)	0.669	0.313	

The composite constraint norm decreased from 0.669 at initialization to 0.313 after the first outer iteration, a reduction of approximately 53%. The dominant contributor to the composite norm is the quasi-symmetry residual c_4, which improved from -0.638 to -0.313. The aspect ratio residual c_0 decreased modestly from $+0.200$ to $+0.170$. The iota constraint c_3 and secondary equilibrium constraints c_2 remained near zero throughout, indicating these were nearly satisfied from initialization.

7.2 Lagrange multiplier and penalty evolution

Table 2 shows the Lagrange multiplier and penalty parameter evolution over the first outer iteration.

Table 2: Lagrange multipliers and penalty parameters at initialization and after outer iteration 1.

	λ_i		μ_i	
Constraint	Init	Iter. 1	Init	Iter. 1
c_0	-0.417	-17.37	10	100
c_1	0.000	0.000	10	10
c_2	-1.14×10^{-4}	$+0.0103$	10	100
c_3	-0.302	-0.287	10	100
c_4	$+0.147$	$+31.48$	10	100

The large magnitude increase in λ_0 (aspect ratio) and λ_4 (quasi-symmetry) reflects the AL method's response to persistent constraint violation: because neither c_0 nor c_4 were satisfied within tolerance, the penalty parameters for both were escalated from $\mu = 10$ to $\mu = 100$ (factor $\tau = 10$), and the

multipliers were updated accordingly. The constraint c_1 remained trivially satisfied (residual exactly zero) and thus its penalty remained at 10. This selective escalation behavior is a key feature of the AL method: it focuses computational pressure on the most problematic constraints while leaving well-satisfied ones undisturbed.

7.3 Solver metrics

The inner L-BFGS-B solver terminated after 5 iterations (the specified limit) with a gradient norm of 67.4 and a wall-clock time of 243 seconds for the first outer iteration. The solver status indicates that the iteration limit was reached before gradient convergence, suggesting that additional inner iterations would yield further constraint reduction before the outer multiplier update. The decision variable displacement $\|x^{k+1} - x^k\|$ was 0.048, indicating a meaningful but not extreme step in the design space.

The initialization phase completed in 0.54 seconds, confirming that the overhead of the AL framework itself is negligible relative to the cost of VMEC evaluations.

7.4 Convergence trajectory

Figure 1 shows the constraint residual history across the first Fourier continuation stage. The quasi-symmetry residual c_4 exhibits the most significant change, consistent with its role as the dominant physics objective. The aspect ratio constraint c_0 shows a smaller but meaningful reduction. The near-zero constraints c_1, c_2, c_3 remain flat, indicating they are passive during this stage.

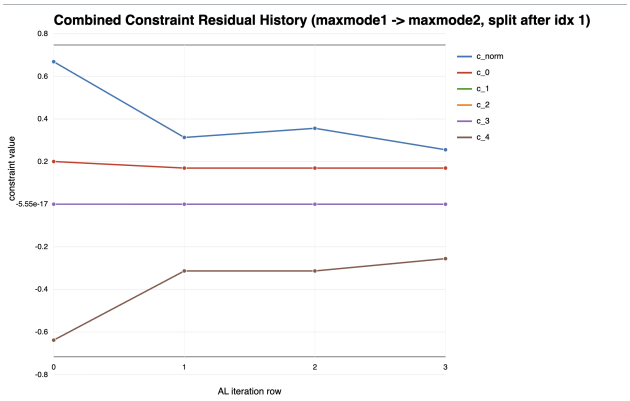


Figure 1: Constraint Residual History

8 Discussion

8.1 Single-stage optimization: benefits realized

The results from the first Fourier continuation stage demonstrate several concrete benefits of the single-stage AL approach. First, no manual penalty weight tuning was required: the initial penalty

parameters $\mu_i = 10$ were set uniformly, and the adaptive escalation scheme automatically identified the quasi-symmetry and aspect ratio constraints as the most problematic and increased their penalties. This contrasts with the two-stage baseline, where appropriate penalty weights for the QH residual objective must be determined through extensive prior experience or parameter sweeps.

Second, the AL framework provides interpretable diagnostic information. The Lagrange multiplier values at convergence have a direct physical interpretation as constraint sensitivity coefficients: the large magnitude of $\lambda_4 = 31.5$ indicates that the quasi-symmetry constraint is strongly active and that the optimizer is allocating significant effort toward its enforcement. This interpretability is absent in penalty-only formulations.

Third, the simultaneous optimization of boundary and coil geometry avoids the structural coupling mismatch of the two-stage approach. In two-stage optimization, the stage-one boundary target may be coil-infeasible, requiring manual relaxation of physics requirements. In the single-stage framework, this tradeoff is explored continuously within a single optimization loop.

8.2 Limitations and open challenges

Several important limitations must be acknowledged. The current results represent only the first outer AL iteration of the first Fourier continuation stage. The inner L-BFGS-B solver reached its iteration limit without achieving gradient convergence, indicating that more inner iterations are needed per outer step. The large gradient norm of 67.4 at termination suggests that the inner minimization was far from a stationary point, which means the multiplier update at this iteration is premature by classical AL theory [12]. Subsequent stages with `max_mode` $\in \{2, 3\}$ are needed to assess whether the FC strategy achieves the anticipated improvement in physics accuracy.

The gradient-free inner minimization via finite differences is computationally expensive. Each gradient evaluation requires $2n$ VMEC calls for n decision variables, making the cost of the inner loop scale as $O(n)$ per gradient step. For `max_mode` $= 3$, the number of boundary modes grows to several hundred, potentially making this approach computationally prohibitive without access to analytic or adjoint gradients [8].

Finally, the comparison with the two-stage baseline requires completion of the full FC optimization pipeline. The current snapshot represents an early stage of a longer computation.

8.3 Implications for fusion energy design

From a broader perspective, the development of more principled, automated optimization frameworks for stellarators has direct implications for the speed and cost of fusion reactor design. The manual iteration cycles characteristic of current two-stage workflows represent a significant bottleneck in the stellarator design process, limiting the number of configurations that can be explored in a given development timeline [1]. Single-stage frameworks with robust constraint handling could dramatically expand the design space that can be explored systematically, potentially identifying configurations with superior confinement properties at lower engineering cost.

9 Conclusion and Future Work

We have presented a single-stage stellarator optimization framework that unifies plasma boundary and coil design using an Augmented Lagrangian formulation combined with a Fourier Continuation spectral conditioning strategy. The framework is implemented within the SIMSOPT ecosystem and targets quasi-helically symmetric configurations with four field periods. Preliminary results from the first FC stage demonstrate a 53% reduction in the composite constraint norm over a single outer AL iteration, with the quasi-symmetry residual improving from -0.638 to -0.313 and the aspect ratio residual decreasing from $+0.200$ to $+0.170$.

The AL formulation provides principled constraint enforcement without manual penalty tuning, interpretable Lagrange multiplier diagnostics, and a unified optimization loop that avoids the structural coupling limitations of two-stage approaches. The Fourier Continuation strategy addresses the ill-conditioning of Fourier boundary parameterizations by progressively expanding the active mode set, warm-starting each stage.

Several important directions remain for future work. First, the full FC pipeline (`max_mode` $\in \{1, 2, 3\}$) must be run to completion to enable meaningful comparison with the two-stage baseline. Second, integration of adjoint gradients through VMEC or DESC would dramatically reduce the cost of the inner minimization and make the framework tractable for higher-resolution cases. Third, extension to quasi-isodynamic configurations and investigation of global optimization strategies within the AL-conditioned space represent important longer-term research directions. Finally, reactor-scale geometries with $O(10^3)$ degrees of freedom and coil feasibility constraints derived from materials and manufacturing considerations will require further algorithmic development and high-performance computing infrastructure.

This work contributes a foundational framework and preliminary evidence that single-stage stellarator optimization with Augmented Lagrangians is a viable and promising direction for the field. As fusion energy development accelerates globally, the ability to explore stellarator design spaces more efficiently and systematically will be a key enabling capability.

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